1. Application model - fit on training split

Model Information	
Data Set	WORK.PDZ
Response Variable	y
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	311589
Number of Observations Used	311589

Response Profile

Ordered Value	y	Total Frequency
1	1	12047
2	0	299542

Probability modeled is y='1'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics

Criterion	Intercept Only	Intercept and Covariates
AIC	101998.74	97824.869
SC	102009.39	97963.311
-2 Log L	101996.74	97798.869

Testing Global Null Hypothesis: BETA=0

Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	4197.8692	12	<.0001
Score	4056.6391	12	<.0001
Wald	3899.8882	12	<.0001

Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-3.6230	0.0477	5763.9427	<.0001
score	1	-0.0839	0.0105	63.8694	<.0001
orig_apr	1	0.4611	0.0134	1183.5611	<.0001
orig_term	1	-0.0882	0.0108	66.8445	<.0001
pti	1	0.1835	0.00992	341.8983	<.0001
ltv	1	0.2135	0.0122	305.8678	<.0001
log_amount	1	0.0724	0.0129	31.4807	<.0001
age_months	1	0.1218	0.0113	115.7265	<.0001

1. Application model - fit on training split

Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
used	1	0.1706	0.0453	14.1536	0.0002
income_verified_3	1	0.0100	0.0298	0.1126	0.7371
employment_verified_	1	0.1052	0.0267	15.5312	<.0001
subvented_any	1	0.1688	0.0468	12.9945	0.0003
score_missing	1	-0.0912	0.1183	0.5938	0.4410

Odds Ratio Estimates

Effect	Point Estimate	95% Wald Confidence Limits	
score	0.920	0.901	0.939
orig_apr	1.586	1.545	1.628
orig_term	0.916	0.896	0.935
pti	1.201	1.178	1.225
ltv	1.238	1.209	1.268
log_amount	1.075	1.048	1.103
age_months	1.129	1.105	1.155
used	1.186	1.085	1.296
income_verified_3	1.010	0.953	1.071
employment_verified_	1.111	1.054	1.171
subvented_any	1.184	1.080	1.298
score_missing	0.913	0.724	1.151

Association of Predicted Probabilities and Observed Responses

Percent Concordant	67.0	Somers' D	0.340
Percent Discordant	33.0	Gamma	0.340
Percent Tied	0.0	Tau-a	0.025
Pairs	3608582474	c	0.670

1. Application model - test and out-of-time AUC (see Fit Statistics: AUC)

Fit Statistics for SCORE Data

Data Set	Total Frequency	Log Likelihood	Error Rate	AIC	AICC	BIC	SC	R-Square
WORK.PDZ	133704	-21104.8	0.0387	42235.69	42235.69	42363.13	42363.13	0.011704
WORK.PDZ	48103	-9402.1	0.0509	18830.26	18830.26	18944.41	18944.41	0.011159
WORK.PDZ	40722	-8006.4	0.0517	16038.76	16038.76	16150.74	16150.74	0.013902

Data Set	Max-Rescaled R-Square	AUC	Brier Score
WORK.PDZ	0.041912	0.658109	0.036738
WORK.PDZ	0.033701	0.652458	0.047763
WORK.PDZ	0.041559	0.660969	0.048371

2. Behavioral model - fit on training split

Model Information	
Data Set	WORK.PDZ
Response Variable	y
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	311589
Number of Observations Used	311589

Response Profile

Ordered Value	y	Total Frequency
1	1	12047
2	0	299542

Probability modeled is y='1'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics

Criterion	Intercept Only	Intercept and Covariates
AIC	101998.74	76348.944
SC	102009.39	76551.284
-2 Log L	101996.74	76310.944

Testing Global Null Hypothesis: BETA=0

Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	25685.7935	18	<.0001
Score	51867.8314	18	<.0001
Wald	22444.2889	18	<.0001

Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-4.5902	0.0535	7360.8202	<.0001
score	1	0.0510	0.0115	19.7671	<.0001
orig_apr	1	0.3296	0.0146	509.3761	<.0001
orig_term	1	-0.0697	0.0119	34.5387	<.0001
pti	1	0.1633	0.0109	225.3244	<.0001
ltv	1	0.2509	0.0134	348.9305	<.0001
log_amount	1	-0.00430	0.0143	0.0903	0.7638
age_months	1	-0.0298	0.0129	5.3456	0.0208

2. Behavioral model - fit on training split

Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
used	1	0.1321	0.0491	7.2455	0.0071
income_verified_3	1	-0.0604	0.0322	3.5130	0.0609
employment_verified_	1	0.0637	0.0288	4.8815	0.0271
subvented_any	1	0.1863	0.0508	13.4197	0.0002
score_missing	1	-0.0557	0.1323	0.1771	0.6739
dq30_0_2m	1	2.1917	0.0267	6720.2971	<.0001
dq30_3_5m	1	1.5734	0.0323	2373.3447	<.0001
dq30_6m_plus	1	1.2844	0.0443	842.1465	<.0001
dq60_0_2m	1	1.6170	0.0292	3075.7296	<.0001
dq60_3m_plus	1	0.9976	0.0469	452.9629	<.0001
dq90_ever	1	1.6740	0.0486	1186.8334	<.0001

Odds Ratio Estimates

Effect	Point Estimate	95% Wald Confidence Limits	
score	1.052	1.029	1.076
orig_apr	1.390	1.351	1.431
orig_term	0.933	0.911	0.955
pti	1.177	1.153	1.203
ltv	1.285	1.252	1.319
log_amount	0.996	0.968	1.024
age_months	0.971	0.946	0.995
used	1.141	1.037	1.256
income_verified_3	0.941	0.884	1.003
employment_verified_	1.066	1.007	1.128
subvented_any	1.205	1.090	1.331
score_missing	0.946	0.730	1.226
dq30_0_2m	8.951	8.494	9.432
dq30_3_5m	4.823	4.527	5.138
dq30_6m_plus	3.612	3.312	3.940
dq60_0_2m	5.038	4.758	5.334
dq60_3m_plus	2.712	2.474	2.973
dq90_ever	5.334	4.849	5.867

Association of Predicted Probabilities and Observed Responses

Percent Concordant	84.8	Somers' D	0.697
Percent Discordant	15.2	Gamma	0.697
Percent Tied	0.0	Tau-a	0.052
Pairs	3608582474	c	0.848

2. Behavioral model - test and out-of-time AUC

Fit Statistics for SCORE Data

Data Set	Total Frequency	Log Likelihood	Error Rate	AIC	AICC	BIC	SC	R-Square
WORK.PDZ	133704	-16394.7	0.0364	32827.35	32827.35	33013.61	33013.61	0.07894
WORK.PDZ	48103	-7457.5	0.0485	14953	14953.01	15119.84	15119.84	0.087963
WORK.PDZ	40722	-6485.9	0.0510	13009.78	13009.8	13173.45	13173.45	0.084858

Data Set	Max-Rescaled R-Square	AUC	Brier Score
WORK.PDZ	0.282681	0.844565	0.030463
WORK.PDZ	0.265658	0.842709	0.040521
WORK.PDZ	0.253683	0.840174	0.042147

2. Behavioral model - confidence intervals clustered by loan (loans repeat across snapshots)

Model Information	
Data Set	WORK.PDZ
Response Variable	y
Number of Response Levels	2
Cluster Variable	asset_id
Number of Clusters	48185
Model	Binary Logit
Optimization Technique	Fisher's Scoring
Variance Adjustment	Degrees of Freedom (DF)

Variance Estimation	
Method	Taylor Series
Variance Adjustment	Degrees of Freedom (DF)

Number of Observations Read	311589
Number of Observations Used	311589

Response Profile

Ordered Value	*y*	*Total Frequency*
1	0	299542
2	1	12047

Probability modeled is y='1'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics

Criterion	*Intercept Only*	*Intercept and Covariates*
AIC	101998.74	76348.944
SC	102009.39	76551.284
-2 Log L	101996.74	76310.944

Testing Global Null Hypothesis: BETA=0

Test	*F Value*	*Num DF*	*Den DF*	*Pr > F*
Likelihood Ratio	406.25	16.0758	774594	<.0001
Score	655.26	18	48167	<.0001
Wald	647.28	18	48167	<.0001

NOTE:
Second-order Rao-Scott design correction 0.1197 applied to the Likelihood Ratio test.

2. Behavioral model - confidence intervals clustered by loan (loans repeat across snapshots)

Analysis of Maximum Likelihood Estimates

Parameter	Estimate	Standard Error	t Value	Pr > |t|
Intercept	-4.5902	0.1086	-42.25	<.0001
score	0.0510	0.0241	2.12	0.0342
orig_apr	0.3296	0.0302	10.92	<.0001
orig_term	-0.0697	0.0238	-2.92	0.0035
pti	0.1633	0.0240	6.80	<.0001
ltv	0.2509	0.0278	9.04	<.0001
log_amount	-0.00430	0.0297	-0.14	0.8849
age_months	-0.0298	0.0260	-1.15	0.2513
used	0.1321	0.0996	1.33	0.1845
income_verified_3	-0.0604	0.0678	-0.89	0.3735
employment_verified_	0.0637	0.0615	1.04	0.3002
subvented_any	0.1863	0.1034	1.80	0.0717
score_missing	-0.0557	0.2675	-0.21	0.8351
dq30_0_2m	2.1917	0.0338	64.87	<.0001
dq30_3_5m	1.5734	0.0526	29.93	<.0001
dq30_6m_plus	1.2844	0.0704	18.25	<.0001
dq60_0_2m	1.6170	0.0383	42.19	<.0001
dq60_3m_plus	0.9976	0.0795	12.55	<.0001
dq90_ever	1.6740	0.0777	21.55	<.0001

NOTE: The degrees of freedom for the t tests is 48184.

Odds Ratio Estimates

Effect	Point Estimate	95% Confidence Limits	
score	1.052	1.004	1.103
orig_apr	1.390	1.311	1.475
orig_term	0.933	0.890	0.977
pti	1.177	1.123	1.234
ltv	1.285	1.217	1.357
log_amount	0.996	0.939	1.055
age_months	0.971	0.922	1.021
used	1.141	0.939	1.387
income_verified_3	0.941	0.824	1.075
employment_verified_	1.066	0.945	1.202
subvented_any	1.205	0.984	1.476
score_missing	0.946	0.560	1.598
dq30_0_2m	8.951	8.377	9.563
dq30_3_5m	4.823	4.351	5.346
dq30_6m_plus	3.612	3.147	4.147

NOTE:
The degrees of freedom in computing the confidence limits is 48184.

2. Behavioral model - confidence intervals clustered by loan (loans repeat across snapshots)

Odds Ratio Estimates

Effect	*Point Estimate*	*95% Confidence Limits*	
dq60_0_2m	5.038	4.673	5.431
dq60_3m_plus	2.712	2.321	3.169
dq90_ever	5.334	4.580	6.211

NOTE: The degrees of freedom in computing the confidence limits is 48184.

Association of Predicted Probabilities and Observed Responses

Percent Concordant	83.9	*Somers' D*	0.696
Percent Discordant	14.3	*Gamma*	0.709
Percent Tied	1.8	*Tau-a*	0.052
Pairs	3608582474	*c*	0.848

2. Behavioral model - out-of-time calibration by decile (Jan-2026)

Rank for Variable P_1	N Obs	Variable	Label	Mean	N
0	4072	P_1	Predicted Probability: y=1	0.0053	4072
		y		0.0020	4072
1	4072	P_1	Predicted Probability: y=1	0.0080	4072
		y		0.0064	4072
2	4072	P_1	Predicted Probability: y=1	0.0103	4072
		y		0.0113	4072
3	4073	P_1	Predicted Probability: y=1	0.0129	4073
		y		0.0098	4073
4	4072	P_1	Predicted Probability: y=1	0.0162	4072
		y		0.0172	4072
5	4072	P_1	Predicted Probability: y=1	0.0213	4072
		y		0.0206	4072
6	4073	P_1	Predicted Probability: y=1	0.0323	4073
		y		0.0358	4073
7	4072	P_1	Predicted Probability: y=1	0.0545	4072
		y		0.0604	4072
8	4072	P_1	Predicted Probability: y=1	0.0962	4072
		y		0.0990	4072
9	4072	P_1	Predicted Probability: y=1	0.2742	4072
		y		0.2549	4072

3. Principal components

Observations	311589
Variables	7

Simple Statistics

	score	orig_apr	orig_term	pti	ltv	log_amount	age_months
Mean	584.6832847	0.1628323811	74.79677716	0.0971404186	1.064257733	10.15102364	23.32147797
StD	49.7983196	0.0432837124	6.73075280	0.0383704517	0.200427746	0.41945062	21.81527210

Correlation Matrix

	score	orig_apr	orig_term	pti	ltv	log_amount	age_months
score	1.0000	-.2711	0.0438	0.0312	0.1223	0.2032	-.0663
orig_apr	-.2711	1.0000	-.1139	0.1392	0.2846	-.2879	-.3721
orig_term	0.0438	-.1139	1.0000	0.1449	0.0459	0.4324	0.0447
pti	0.0312	0.1392	0.1449	1.0000	0.0818	0.1728	-.0302
ltv	0.1223	0.2846	0.0459	0.0818	1.0000	0.1049	0.0102
log_amount	0.2032	-.2879	0.4324	0.1728	0.1049	1.0000	-.0925
age_months	-.0663	-.3721	0.0447	-.0302	0.0102	-.0925	1.0000

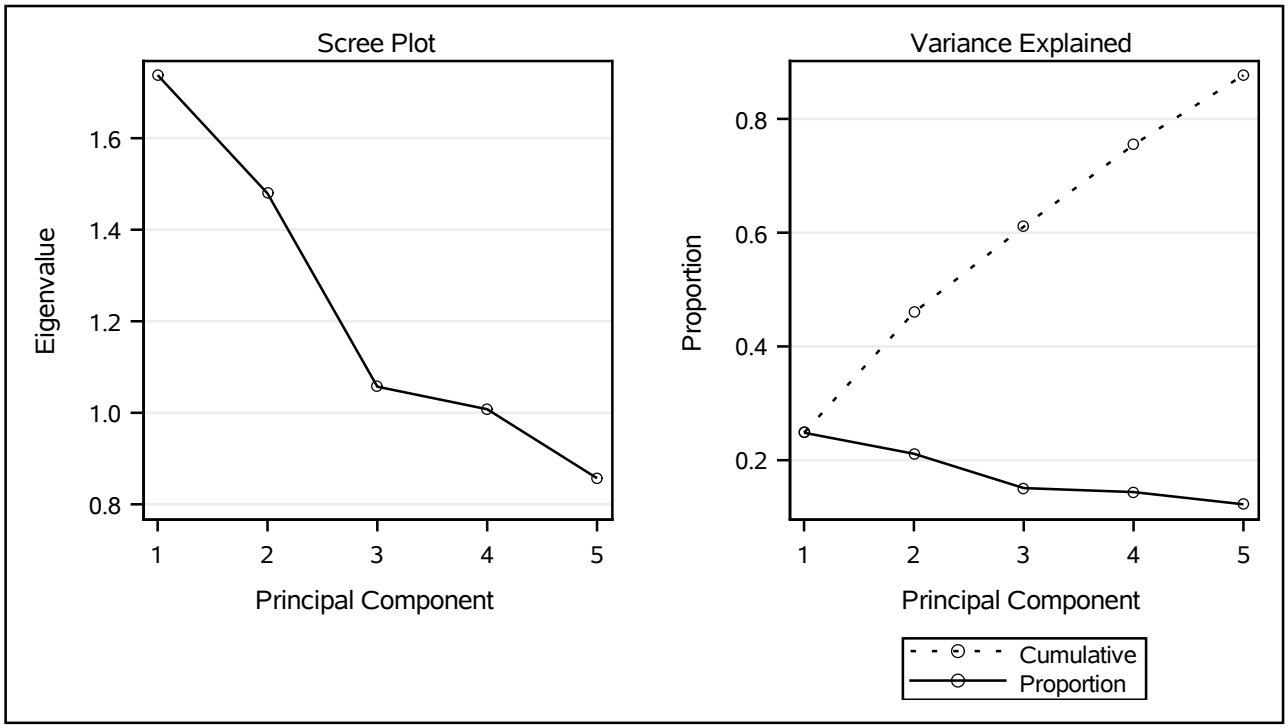
Eigenvalues of the Correlation Matrix

	Eigenvalue	Difference	Proportion	Cumulative
1	1.73782901	0.25855822	0.2483	0.2483
2	1.47927079	0.42226714	0.2113	0.4596
3	1.05700365	0.04919228	0.1510	0.6106
4	1.00781136	0.15077290	0.1440	0.7546
5	0.85703847		0.1224	0.8770

Eigenvectors

	Prin1	Prin2	Prin3	Prin4	Prin5
score	0.363070	0.036340	-.735689	0.079309	0.296534
orig_apr	-.490906	0.529057	0.085815	0.018355	-.093487
orig_term	0.481538	0.231142	0.393391	-.084917	-.376647
pti	0.148703	0.423311	0.353626	0.070692	0.802090
ltv	0.003144	0.462404	-.229184	0.708570	-.281015
log_amount	0.588378	0.248520	0.019247	-.172956	-.178043
age_months	0.164322	-.458889	0.344519	0.670209	0.086299

3. Principal components



4a. Cox model, constant effects - expected APR hazard ratio 1.680

Model Information	
Data Set	WORK.SURVZ
Entry Time Variable	entry
Dependent Variable	time
Censoring Variable	co
Censoring Value(s)	0
Ties Handling	EFRON

Number of Observations Read	50156
Number of Observations Used	50156

Summary of the Number of Event and Censored Values

Total	*Event*	*Censored*	*Percent Censored*
50156	2929	47227	94.16

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics

Criterion	*Without Covariates*	*With Covariates*
-2 LOG L	54956.965	53848.882
AIC	54956.965	53870.882
SBC	54956.965	53936.688

Testing Global Null Hypothesis: BETA=0

Test	*Chi-Square*	*DF*	*Pr > ChiSq*
Likelihood Ratio	1108.0829	11	<.0001
Score	1107.8615	11	<.0001
Wald	1071.1172	11	<.0001

Analysis of Maximum Likelihood Estimates

Parameter	*DF*	*Parameter Estimate*	*Standard Error*	*Chi-Square*	*Pr > ChiSq*	*Hazard Ratio*	*95% Hazard Ratio Confidence Limits*	
score	1	-0.05505	0.02081	6.9980	0.0082	0.946	0.909	0.986
orig_apr	1	0.51881	0.02695	370.6121	<.0001	1.680	1.594	1.771
orig_term	1	-0.10641	0.02036	27.3139	<.0001	0.899	0.864	0.936
pti	1	0.15038	0.02002	56.4150	<.0001	1.162	1.118	1.209
ltv	1	0.20972	0.02465	72.3765	<.0001	1.233	1.175	1.294
log_amount	1	0.02169	0.02578	0.7083	0.4000	1.022	0.972	1.075
used	1	0.04971	0.09157	0.2947	0.5872	1.051	0.878	1.258

4a. Cox model, constant effects - expected APR hazard ratio 1.680

Analysis of Maximum Likelihood Estimates

Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits	
income_verified_3	1	-0.00190	0.05947	0.0010	0.9745	0.998	0.888	1.121
employment_verified_	1	0.06578	0.05246	1.5726	0.2098	1.068	0.964	1.184
subvented_any	1	0.14506	0.09469	2.3469	0.1255	1.156	0.960	1.392
score_missing	1	-0.21407	0.24373	0.7714	0.3798	0.807	0.501	1.302

4a. Proportional-hazards check - expected correlation for APR about -0.10

1 With Variables:	time
11 Variables:	r_score r_apr r_term r_pti r_ltv r_amt r_used r_inc r_emp r_sub r_miss

Simple Statistics

Variable	N	Mean	Std Dev	Sum	Minimum	Maximum	Label
time	2929	23.87743	19.86712	69937	4.00000	87.00000	
r_score	2929	1.19119E-8	1.03758	0.0000349	-3.74513	4.15662	Schoenfeld Residual for score
r_apr	2929	3.69233E-7	0.86422	0.00108	-3.37537	2.48334	Schoenfeld Residual for orig_apr
r_term	2929	1.2148E-6	0.92274	0.00356	-4.41363	1.90349	Schoenfeld Residual for orig_term
r_pti	2929	3.56163E-7	1.01816	0.00104	-2.70632	16.94726	Schoenfeld Residual for pti
r_ltv	2929	9.47602E-7	0.80217	0.00278	-4.40683	1.92380	Schoenfeld Residual for ltv
r_amt	2929	5.81054E-7	0.89594	0.00170	-2.68680	3.07851	Schoenfeld Residual for log_amount
r_used	2929	4.95953E-8	0.37147	0.0001453	-0.87020	0.57952	Schoenfeld Residual for used
r_inc	2929	2.14292E-7	0.43503	0.0006277	-0.83748	0.33727	Schoenfeld Residual for income_verified_3
r_emp	2929	1.74196E-7	0.45318	0.0005102	-0.80971	0.39490	Schoenfeld Residual for employment_verified_3
r_sub	2929	-5.3566E-8	0.35224	-0.0001569	-0.55955	0.90778	Schoenfeld Residual for subvented_any
r_miss	2929	9.3616E-10	0.07596	2.74203E-6	-0.00974	0.99454	Schoenfeld Residual for score_missing

Pearson Correlation Coefficients, N = 2929

	r_score	r_apr	r_term	r_pti	r_ltv	r_amt	r_used	r_inc	r_emp	r_sub	r_miss
time	0.01626	-0.10261	-0.04417	-0.03345	-0.06376	-0.00527	-0.03939	-0.01323	-0.01659	0.03321	-0.02571

4b. Cox model, APR effect changes after age 24 - expected APR HR 1.784, apr_gt24 HR 0.809

Model Information	
Data Set	WORK.SURVZ
Entry Time Variable	entry
Dependent Variable	time
Censoring Variable	co
Censoring Value(s)	0
Ties Handling	EFRON

Number of Observations Read	50156
Number of Observations Used	50156

Summary of the Number of Event and Censored Values

Total	*Event*	*Censored*	*Percent Censored*
50156	2929	47227	94.16

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics

Criterion	*Without Covariates*	*With Covariates*
-2 LOG L	54956.965	53827.387
AIC	54956.965	53851.387
SBC	54956.965	53923.176

Testing Global Null Hypothesis: BETA=0

Test	*Chi-Square*	*DF*	*Pr > ChiSq*
Likelihood Ratio	1129.5772	12	<.0001
Score	1128.0753	12	<.0001
Wald	1085.8674	12	<.0001

Analysis of Maximum Likelihood Estimates

Parameter	*DF*	*Parameter Estimate*	*Standard Error*	*Chi-Square*	*Pr > ChiSq*	*Hazard Ratio*	*95% Hazard Ratio Confidence Limits*	
score	1	-0.05553	0.02079	7.1340	0.0076	0.946	0.908	0.985
orig_apr	1	0.57905	0.02985	376.2246	<.0001	1.784	1.683	1.892
orig_term	1	-0.10621	0.02035	27.2267	<.0001	0.899	0.864	0.936
pti	1	0.14769	0.02002	54.4439	<.0001	1.159	1.115	1.206
ltv	1	0.20405	0.02475	67.9928	<.0001	1.226	1.168	1.287
log_amount	1	0.02468	0.02580	0.9151	0.3388	1.025	0.974	1.078
used	1	0.05007	0.09233	0.2941	0.5876	1.051	0.877	1.260

4b. Cox model, APR effect changes after age 24 - expected APR HR 1.784, apr_gt24 HR 0.809

Analysis of Maximum Likelihood Estimates

Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits	
income_verified_3	1	-0.00816	0.05942	0.0189	0.8907	0.992	0.883	1.114
employment_verified_	1	0.06516	0.05237	1.5481	0.2134	1.067	0.963	1.183
subvented_any	1	0.11263	0.09599	1.3768	0.2406	1.119	0.927	1.351
score_missing	1	-0.20657	0.24372	0.7183	0.3967	0.813	0.504	1.311
apr_gt24	1	-0.21214	0.04589	21.3688	<.0001	0.809	0.739	0.885